Exelon Generation.

Clinton Power Station
8401 Power Road
Clinton, IL 61727

U-604608

10 CFR 50.73
SRRS 5A.108

February 22, 2021

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D.C. 20555-0001

Clinton Power Station, Unit 1
Facility Operating License No. NPF-62
NRC Docket No. 50-461

Subject: Licensee Event Report 2020-003-00

Enclosed is Licensee Event Report (LER) 2020-003-00: Reactor Core Isolation Cooling System Card Fuse Failure Results in a Condition Prohibited by Technical Specifications and That Could Have Prevented Fulfillment of a Safety Function Needed to Control the Release of Radioactive Material. This report is being submitted in accordance with the requirements of 10 CFR 50.73.

There are no regulatory commitments contained in this report.

Should you have any questions concerning this report, please contact Mr. Dale Shelton, Regulatory Assurance Manager, at (217) 937-2800.

Respectfully,

Thomas D. Chalmers
Site Vice President
Clinton Power Station

Attachment: Licensee Event Report 2020-003-00

cc:

Regional Administrator - Region III
NRC Senior Resident Inspector - Clinton Power Station
Office of Nuclear Facility Safety - Illinois Emergency Management Agency

NRC FORM 366 (08-2020)

U.S. NUCLEAR REGULATORY COMMISSION

LICENSEE EVENT REPORT (LER)

(See Page 3 for required number of digits/characters for each block)
(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name	2. Docket Number	3. Page
Clinton Power Station, Unit 1	05000461	1 OF 3

4. Title

Reactor Core Isolation Cooling System Card Fuse Failure Results in a Condition Prohibited by Technical Specifications and That Could Have Prevented Fulfillment of a Safety Function Needed to Control the Release of Radioactive Material

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number
12	26	2020	2020	- 003 -	00	02	22	2021		05000
									Facility Name	Docket Number 05000

9. Operating Mode	10. Power Level
1	098

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(4)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(i)(A)	☒ 50.73(a)(2)(v)(C)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(1)(i)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(iii)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(iv)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	
☐ 20.2203(a)(2)(v)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	

☐ **OTHER** (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Dale Shelton, Regulatory Assurance Manager	(217) 937-2800

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
XX	BN	FU	GE	Y					

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

On 12/26/20, at 0205 CST, during performance of Reactor Core Isolation Cooling (RCIC) surveillance procedures, the Division 2 RCIC indication response was not as expected. Troubleshooting determined the indication response was due to failure of the Div. 2 RCIC-5 card, which provides an isolation signal. Technical Specification (TS) 3.3.6.1, Primary Containment and Drywell Isolation Instrumentation, was entered. Past operability review found that the condition existed from 12/21/20, through 12/26/20, exceeding the TS 3.3.6.1, Action D.1, 24-hour action statement. The Div. 2 RCIC-5 card failed due to an internal blown fuse. The cause of the missed opportunity to identify the failure on 12/21/20 was Operations did not properly recognize the risk associated with the condition and initiate timely actions to evaluate the impact on operability. The card was replaced and tested on 12/26/20, at 1506 CST. Refresher training will be conducted for appropriate Operations personnel regarding the need to perform manual self tests or use alternate indications when anomalous indications or alarms occur. This condition is reportable under 10 CFR 50.73(a)(2)(i)(B) and 10 CFR 50.73(a)(2)(v)(C). There was no impact to the health and safety of the public or plant personnel from this condition as the isolation safety function was maintained throughout the event. This event does not meet the criteria for a Safety System Functional Failure.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
Clinton Power Station, Unit 1	05000461	2020	- 003	- 00

NARRATIVE

PLANT AND SYSTEM IDENTIFICATION

General Electric -- Boiling Water Reactor, 3473 Megawatts Thermal Rated Core Power
Energy Industry Identification System (EIIS) codes are identified in text as [XX].

EVENT IDENTIFICATION

Reactor Core Isolation Cooling System Card Fuse Failure Results a Condition Prohibited by Technical Specifications and That Could Have Prevented Fulfillment of a Safety Function Needed to Control the Release of Radioactive Material

A. Plant Operating Conditions Before the Event

Unit: 1	Event Date: December 26, 2020	Event Time: 0205
Mode: 1	Mode Name: Power Operation	Reactor Power: 098

B. Description of Event

On December 26, 2020, at approximately 0205 CST, during performance of Reactor Core Isolation Cooling (RCIC) system [BN] steam line flow and supply pressure surveillance procedures 9030.01C034 and 9030.01C035, the Division 2 RCIC indication response was not as expected. Technical Specification (TS) 3.3.6.1, Primary Containment and Drywell Isolation Instrumentation, Action D.1, place affected channel in trip within 24 hours, was entered. Troubleshooting determined the indication response was due to a failure of the Division 2 RCIC-5 card, D-A14-A112, which was not sending an isolation signal during the surveillance. The failed Division 2 RCIC-5 card was replaced and successfully tested on December 26, 2020, at 1506 CST and TS 3.3.6.1, Action D.1, was exited.

The failed Division 2 RCIC-5 card provides an isolation signal that automatically initiates closure of appropriate Primary Containment Isolation Valves (PCIVs) and drywell isolation valves.

A review for past operability found that the condition could have been identified during performance of procedure 9030.01C041, RCIC Turbine Exhaust Diaphragm Pressure, on December 21, 2020, at 1515 CST, when an alarm did not occur as expected. Although the alarm was not required in order to meet TS surveillance requirements based on the procedure acceptance criteria, further investigation determined the failure was an indication the Division 2 RCIC-5 card was not providing an output isolation signal. Therefore, there is firm evidence that the condition existed from December 21, 2020, through December 26, 2020, which exceeded the TS 3.3.6.1, Action D.1, 24-hour action statement.

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Clinton Power Station, Unit 1	05000461	2020	- 003	- 00

NARRATIVE

C. Cause of the Event

An analysis of the failed Division 2 RCIC-5 card, D-A14-A112, identified a blown fuse [FU] internal to the card.

The cause of the failure to identify the failed card on December 21, 2020, was Operations did not recognize the risk associated with the condition and did not initiate actions to evaluate the operability.

D. Safety Consequences

The condition described in this LER is reportable under 10 CFR 50.73(a)(2)(i)(B), any operation or condition which was prohibited by the plant's Technical Specifications. Because this condition affected one division of RCIC logic, the safety function to automatically initiate closure of appropriate PCIVs and drywell isolation valves was maintained and would have initiated if required. However, during performance of surveillance procedures 9030.01C034 and 9030.01C035 on December 26, 2020, the redundant automatic isolation capability was removed from service, creating a short duration loss of containment isolation function. This condition is reportable under 10 CFR 50.73(a)(2)(v)(C) as a condition that could have prevented the fulfillment of the safety function of a system needed to control the release of radioactive material. Evaluation of the impact of this leakage pathway determined that the containment function would be met by Operations personnel using existing procedures to perform manual isolations if automatic isolations did not occur as expected, if an event had occurred. Therefore, there was no impact to the health and safety of the public or plant personnel from this condition. This event does not meet the criteria for a Safety System Functional Failure.

E. Corrective Actions

The failed Division 2 RCIC-5 card was replaced and successfully tested on December 26, 2020, at 1506 CST.

Refresher training will be conducted for appropriate Operations personnel regarding the Nuclear System Protection System (NSPS) self-test system, how it is used to meet surveillance requirements, and the need to perform manual self tests or use alternate indications to verify channel operability when anomalous indications or alarms occur.

F. Previous Similar Occurrences

A review of previous LERs did not identify any events that were similar to the condition described in this LER.

G. Component Failure Data

The Division 2 RCIC card that failed is a NSPS (Essential Logic) RCIC-5 made by GE.